Measure FF-08: optimisation of river engineering structures to reduce erosion

Description

River training structures, such as groynes and guiding walls, are used to modify hydraulic flow patterns and sediment dynamics. Their effectiveness depends on design parameters including length, spacing, width, height, orientation, shape, and material, which collectively influence shear stress distribution and channel morphology. Adapting and optimizing these structures to site-specific river conditions can reduce erosional tendencies in undesired locations and increase sediment retention time in free-flowing river reaches.

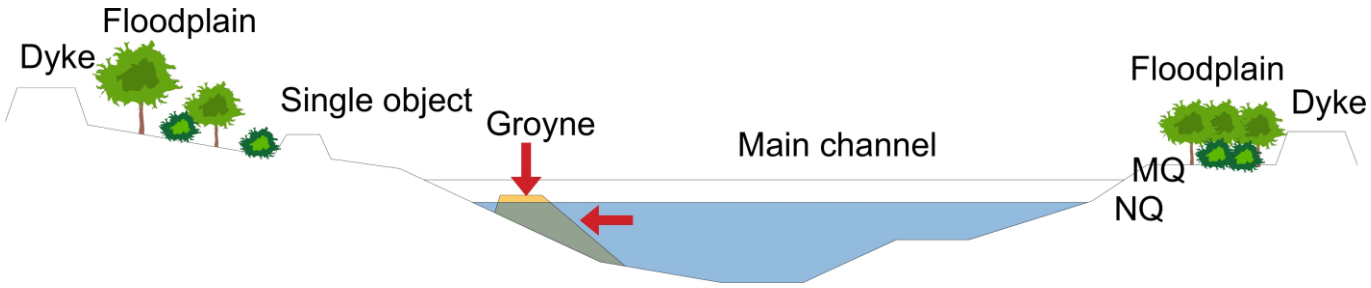
Key management targets reached: 1

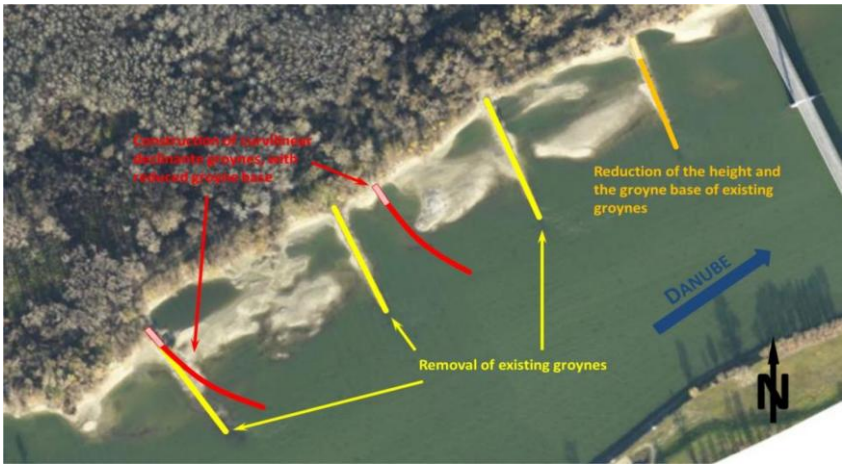
Positive | Neutral | Negative

- Hydromorphological improvement
- Flood risk reduction
- Drought risk reduction
- Ecological and biodiversity improvement
- Sustainable hydropower improvement
- Sustainable navigation improvement

Overview

Aim(s)	Reduce erosion in undesired locations
Location	Free-flowing sections
Spatial scaling	Local, river reach
Temporal scaling	Long term
River classification	Rural channelized rivers





Optimisation of groins for example by:

- shortening
- lowering
- change of orientation
- increase of spacing

Figure based on the EU-projects DanubeSediment (DTP-1-1-195-2.1)

River Connectivity Properties[1]

Positive | Neutral | Negative | Uncertain

Flow	Sediment	Biota	Physical and Chemical Compounds	Nutrients
Longitudinal	Longitudinal	Longitudinal	Longitudinal	Longitudinal
Lateral	Lateral	Lateral	Lateral	Lateral
Vertical	Vertical	Vertical	Vertical	Vertical
Temporal	Temporal	Temporal	Temporal	Temporal

Nature-based Solutions[2]

High | Medium | Low | No

Ecological criteria	Socio-ecological criteria	Socio-economic criteria
Biodiversity	Protection of ecosystems	Climate and disaster mitigation
Resilience	Restoration of ecosystem	Human well-being & health
Ecosystem services	Sustainable management	Livelihoods & economic development

Related Ecosystem Services[3]

High | Medium | Low | Not relevant

Provisioning	Cultural	Regulating	Regulating	Regulating
Wood production		Flood mitigation	Carbon sequestration	Landscape aesthetics
Wild fish production		Low flow mitigation	Carbon retention	Natural and cultural heritage
Grassland production		Local climate cooling	Nitrogen retention	Unspec. Interact. with water bodies
Arable crop production		Sediment regulation	Phosphorous retention	Water related activities
Drinking water provision		Soil formation in floodplains	Maintaining habitats	

Effects of the measure

High | Medium | Low | None

	Parameter	Effects
Hydro-dynamics	Water level	Decreased in main channel, increased in groin field
	Flow velocity	Decreased in main channel, increased variability in groin field
	Shear stress	Decreased in main channel, increased in groin field
Sediment dynamics	Transport capacity	Decreased in main channel, increased in groin field
	Continuity	Eventually decreased in main channel, increased sediment input groin field
Morpho-dynamics	River morphology	Decreased morphodynamics in main channel, increased in groin field
Ecology	Ecological benefits	Improved meso- and micro- habitat diversity

[1] **River Connectivity:** River connectivity refers to the movement of water, sediments, and aquatic life within the riverine network across three spatial and one temporal dimension. River connectivity cannot be adequately assessed solely based on its dimensions. The aspects considered to comprehensively evaluate the biotic and abiotic components are flow, sediment, biota, physical and chemical compounds, and nutrients.

[2] **Nature-based solutions (NbS):** Actions to protect, conserve, restore, sustainably use, and manage natural or modified terrestrial, freshwater, coastal, and marine ecosystems, which address social, economic, and environmental challenges effectively and adaptively, while simultaneously providing human well-being, ecosystem services, resilience, and biodiversity benefits, and recognizes that nature-based solutions [...] respect social and environmental safeguards. (UNEA, 2022).

Specifics

Operation	Not required
Maintenance	No / irregularly
Technology Readiness Level[4]	9
Recommendation to couple with other measures	Riverbank restoration, reconnection of side channels
Probability of reaching desired aim	Very certain
Notes / Risks	Flood protection and inland navigation can pose limits to the extent of the measure / amount of allowable sedimentation

Example

Removal of old groynes and construction of new groynes in the Danube river at Witzelsdorf, Austria
Viadonau, Pilot Project Witzelsdorf, (2023)
https://www.viadonau.org/en/company/project-database/integrated-river-engineering-project-catalogue-of-measures/pilot-project-witzelsdorf

References

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- Liedermann, M., Gmeiner, P. H., Glas, M., Tritthart, M., & Habersack, H. (2016). Funktionalität der getesteten flussbaulichen Maßnahmen im Pilotprojekt Bad Deutsch-Altenburg [Functionality of the tested river engineering measures within the pilot project Bad Deutsch-Altenburg at the Austrian Danube River]. *Österreichische Wasser- und Abfallwirtschaft, 68*(5–6), 217–225 DOI 10.1007/s00506-016-0313-8

[3] **Ecosystem services** are the benefits that humans derive from natural ecosystems. These services are essential for human survival, well-being, and economic activity, as they provide resources, regulate environmental processes, and support cultural and recreational activities. Ecosystem services are often categorized into three types: provisioning, regulating, and cultural services.

[4] **The Technology Readiness Level (TRL)** is a widely used framework used to evaluate the readiness of technologies for commercialization or operational use. The TRL scale typically ranges from 1 to 9, with 1 representing the earliest stage of research and 9 indicating a fully mature and operational technology.